

Linear Approximation and L'Hôpital's Rule

ANSWER KEY



Tangent line approximation

- 1 Use linear approximation to estimate $\sqrt{4.1}$, given $f(x) = \sqrt{x}$, $f(4) = 2$, and $f'(4) = \frac{1}{4}$.

$$L(x) = f(4) + f'(4)(x - 4) = 2 + \frac{1}{4}(x - 4). L(4.1) = 2 + \frac{1}{4}(0.1) = 2.025.$$

- 2 Use linear approximation to estimate $(1.02)^5$, using $f(x) = x^5$ near $x = 1$.

$$f(1) = 1, f'(x) = 5x^4, f'(1) = 5. L(x) = 1 + 5(x - 1). L(1.02) = 1 + 5(0.02) = 1.1.$$

- 3 Find the linearization $L(x)$ of $f(x) = \sin x$ at $x = 0$, and use it to approximate $\sin(0.1)$.

$$f(0) = 0, f'(0) = \cos 0 = 1: L(x) = x. \sin(0.1) \approx L(0.1) = 0.1.$$

Error and over/underestimation

- 4 For $f(x) = \sqrt{x}$ (concave down for $x > 0$), explain whether the linear approximation at $x = 4$ overestimates or underestimates $f(4.1)$.

Since f is concave down, its graph lies below any tangent line, so the tangent line (linear approximation) **OVERESTIMATES** the true value of $f(4.1)$.

- 5 For $f(x) = x^2$ (concave up everywhere), explain whether the tangent line approximation at any point overestimates or underestimates nearby function values.

Since f is concave up, its graph lies above any tangent line, so the tangent line approximation **UNDERESTIMATES** nearby true function values.

Indeterminate forms and L'Hôpital's Rule

- 6 Evaluate $\lim_{x \rightarrow 0} \frac{\sin x}{x}$ using L'Hôpital's Rule, and confirm it matches the known limit.

Form $\frac{0}{0}$: differentiate top and bottom: $\lim_{x \rightarrow 0} \frac{\cos x}{1} = \cos 0 = 1$, matching the well-known limit.

- 7 Evaluate $\lim_{x \rightarrow 0} \frac{e^x - 1 - x}{x^2}$ using L'Hôpital's Rule (applied twice).

Form $\frac{0}{0}$: $\lim_{x \rightarrow 0} \frac{e^x - 1}{2x}$, still $\frac{0}{0}$: apply again: $\lim_{x \rightarrow 0} \frac{e^x}{2} = \frac{1}{2}$.

- 8 Evaluate $\lim_{x \rightarrow \infty} \frac{\ln x}{x}$ using L'Hôpital's Rule.

Form $\frac{\infty}{\infty}$: $\lim_{x \rightarrow \infty} \frac{1/x}{1} = \lim_{x \rightarrow \infty} \frac{1}{x} = 0$.

Recognizing when L'Hôpital's Rule applies

- 9 Explain why L'Hôpital's Rule cannot be directly applied to $\lim_{x \rightarrow 0} \frac{x+1}{x}$, and evaluate the limit by other means.

This is NOT an indeterminate form — as $x \rightarrow 0$, the numerator approaches 1 (not 0) while the denominator approaches 0, so it's a $\frac{1}{0}$ form, not $\frac{0}{0}$. This limit does not exist (it approaches $\pm\infty$ depending on direction), so L'Hôpital's Rule doesn't apply.

Differentials

- 10 For $f(x) = x^3$, find the differential dy when $x = 2$ and $dx = 0.1$, and compare it to the actual change $\Delta y = f(2.1) - f(2)$.

$dy = f'(x) dx = 3x^2 dx = 3(4)(0.1) = 1.2$. Actual: $f(2.1) = 9.261$, $f(2) = 8$, $\Delta y = 1.261$. The differential $dy = 1.2$ closely approximates the actual change $\Delta y = 1.261$.

More L'Hôpital's Rule practice

- 11 Evaluate $\lim_{x \rightarrow 0} \frac{\tan x - x}{x^3}$ using L'Hôpital's Rule (applied as many times as needed).

Form $\frac{0}{0}$: $\lim_{x \rightarrow 0} \frac{\sec^2 x - 1}{3x^2}$, still $\frac{0}{0}$: apply again: $\lim_{x \rightarrow 0} \frac{2\sec^2 x \tan x}{6x}$, still $\frac{0}{0}$: apply once more:
 $\lim_{x \rightarrow 0} \frac{2\sec^2 x (\sec^2 x) + 4\sec^2 x \tan^2 x}{6} = \frac{2(1) + 0}{6} = \frac{1}{3}$.

Visualizing a linear approximation

- 12 The graph shows $f(x) = \sqrt{x}$ together with its tangent line at $x = 4$, $L(x) = 2 + \frac{1}{4}(x - 4)$.

The tangent line closely tracks $f(x) = \sqrt{x}$ near $x = 4$ but lies above the curve further away, visually confirming that the linearization overestimates f (since $\sqrt{x}$ is concave down) — consistent with the earlier conclusion.

L'Hôpital's Rule with exponential indeterminate forms

- 13 Evaluate $\lim_{x \rightarrow \infty} \frac{x^2}{e^x}$ using L'Hôpital's Rule (applied twice).

Form $\frac{\infty}{\infty}$: $\lim_{x \rightarrow \infty} \frac{2x}{e^x}$, still $\frac{\infty}{\infty}$: apply again: $\lim_{x \rightarrow \infty} \frac{2}{e^x} = 0$.

Linear approximation in an applied context

- 14 The side length of a cube is measured as 5 cm, with a possible measurement error of ± 0.02 cm. Use the differential $dV = 3s^2 ds$ to estimate the maximum resulting error in the computed volume $V = s^3$.

$dV = 3(5)^2(0.02) = 3(25)(0.02) = 1.5 \text{ cm}^3$. The volume could be off by as much as approximately 1.5 cm^3 .

A capstone L'Hôpital's Rule problem

- 15 Evaluate $\lim_{x \rightarrow 0^+} x \ln x$ by first rewriting it as a quotient $\frac{\ln x}{1/x}$ to apply L'Hôpital's Rule.

Rewritten as $\frac{\ln x}{1/x}$, this is $\frac{-\infty}{-\infty}$ as $x \rightarrow 0^+$: $\lim_{x \rightarrow 0^+} \frac{1/x}{-1/x^2} = \lim_{x \rightarrow 0^+} (-x) = 0$.

- 16 A surveyor measures the angle of elevation to a tower as $\theta = 60^\circ$ with a possible error of $\pm 1^\circ$ ($\pm \frac{\pi}{180}$ radians). If the tower's height is estimated using $h = 100 \tan \theta$ (with the base 100 m away), use the differential $dh = 100 \sec^2 \theta d\theta$ to estimate the resulting error in the height, to two decimal places.

At $\theta = 60^\circ = \frac{\pi}{3}$: $\sec^2 \theta = 4$. $dh = 100(4)(\frac{\pi}{180}) \approx 400(0.01745) \approx 6.98 \text{ m}$ — the height estimate could be off by roughly 7 meters.

- 17 Evaluate $\lim_{x \rightarrow \infty} x \sin(\frac{1}{x})$ by rewriting it as $\frac{\sin(1/x)}{1/x}$ and applying L'Hôpital's Rule (or recognizing the standard limit form).

Let $u = \frac{1}{x} \rightarrow 0^+$ as $x \rightarrow \infty$: $\lim_{u \rightarrow 0} \frac{\sin u}{u} = 1$ (the standard limit), so $\lim_{x \rightarrow \infty} x \sin(\frac{1}{x}) = 1$.

- 18 For $f(x) = \cos x$ near $x = 0$, complete each step:

a Find the linearization $L(x)$ at $x = 0$. $f(0) = 1$, $f'(0) = -\sin 0 = 0$: $L(x) = 1$.

b Use $L(x)$ to estimate $\cos(0.05)$. $L(0.05) = 1$.

c Compare to the true value $\cos(0.05) \approx 0.99875$, and state whether the linearization over- or under-estimates. The true value is slightly less than 1, so the linearization (which gives exactly 1) OVERESTIMATES — consistent with $\cos x$ being concave down near $x = 0$.

Multiple choice — easy 15

19 Use linear approximation to estimate $f(2.9)$, given $f(3) = 2$ and $f'(3) = -1$.

a. 2.1 ✓

b. 2

c. 1.9

d. 2.2

$$L(x) = 2 - (x - 3), \text{ so } L(2.9) = 2 - 1(-0.1) = 2.1.$$

20 Use linear approximation to estimate $f(4.2)$, given $f(4) = -3$ and $f'(4) = -3$.

a. -2.4

b. -4.2

c. -3

d. -3.6 ✓

$$L(x) = -3 - 3(x - 4), \text{ so } L(4.2) = -3 - 3(0.2) = -3.6.$$

21 Use linear approximation to estimate $f(1.3)$, given $f(1) = -2$ and $f'(1) = -9$.

a. 0.7

b. -7.4

c. -2

d. -4.7 ✓

$$L(x) = -2 - 9(x - 1), \text{ so } L(1.3) = -2 - 9(0.3) = -4.7.$$

22 Use linear approximation to estimate $f(2.2)$, given $f(2) = 3$ and $f'(2) = -8$.

a. 4.6

b. 1.4 ✓

c. -0.2

d. 3

$$L(x) = 3 - 8(x - 2), \text{ so } L(2.2) = 3 - 8(0.2) = 1.4.$$

23 Use linear approximation to estimate $f(1.9)$, given $f(2) = -5$ and $f'(2) = 9$.

a. -5.9 ✓

b. -6.8

c. -5

d. -4.1

$$L(x) = -5 + 9(x - 2), \text{ so } L(1.9) = -5 + 9(-0.1) = -5.9.$$

24 Use linear approximation to estimate $f(1.3)$, given $f(1) = 3$ and $f'(1) = -8$.

a. 0.6 ✓

b. 3

c. 5.4

d. -1.8

$$L(x) = 3 - 8(x - 1), \text{ so } L(1.3) = 3 - 8(0.3) = 0.6.$$

25 Use linear approximation to estimate $f(6.1)$, given $f(6) = 8$ and $f'(6) = 7$.

a. 7.3

b. 8.7 ✓

c. 9.4

d. 8

$$L(x) = 8 + 7(x - 6), \text{ so } L(6.1) = 8 + 7(0.1) = 8.7.$$

26 Use linear approximation to estimate $f(3.3)$, given $f(3) = 7$ and $f'(3) = -7$.

- a. 2.8
- b. 4.9 ✓
- c. 9.1
- d. 7

$$L(x) = 7 - 7(x - 3), \text{ so } L(3.3) = 7 - 7(0.3) = 4.9.$$

27 Use linear approximation to estimate $f(5.5)$, given $f(6) = -1$ and $f'(6) = 1$.

- a. -1.5 ✓
- b. -2
- c. -1
- d. -0.5

$$L(x) = -1 + (x - 6), \text{ so } L(5.5) = -1 + 1(-0.5) = -1.5.$$

28 Use linear approximation to estimate $f(2.3)$, given $f(2) = -4$ and $f'(2) = 3$.

- a. -4.9
- b. -3.1 ✓
- c. -2.2
- d. -4

$$L(x) = -4 + 3(x - 2), \text{ so } L(2.3) = -4 + 3(0.3) = -3.1.$$

29 Use linear approximation to estimate $f(1.5)$, given $f(1) = 6$ and $f'(1) = 5$.

- a. 6
- b. 3.5
- c. 11
- d. 8.5 ✓

$$L(x) = 6 + 5(x - 1), \text{ so } L(1.5) = 6 + 5(0.5) = 8.5.$$

30 Use linear approximation to estimate $f(6.2)$, given $f(6) = 5$ and $f'(6) = 4$.

a. 6.6

b. 5.8 ✓

c. 4.2

d. 5

$$L(x) = 5 + 4(x - 6), \text{ so } L(6.2) = 5 + 4(0.2) = 5.8.$$

31 Use linear approximation to estimate $f(7.5)$, given $f(8) = 7$ and $f'(8) = -9$.

a. 11.5 ✓

b. 7

c. 16

d. 2.5

$$L(x) = 7 - 9(x - 8), \text{ so } L(7.5) = 7 - 9(-0.5) = 11.5.$$

32 Use linear approximation to estimate $f(7.8)$, given $f(8) = -2$ and $f'(8) = 3$.

a. -2.6 ✓

b. -3.2

c. -2

d. -1.4

$$L(x) = -2 + 3(x - 8), \text{ so } L(7.8) = -2 + 3(-0.2) = -2.6.$$

33 Use linear approximation to estimate $f(4.8)$, given $f(5) = 6$ and $f'(5) = -9$.

- a. 9.6
- b. 4.2
- c. 6
- d. 7.8 ✓

$$L(x) = 6 - 9(x - 5), \text{ so } L(4.8) = 6 - 9(-0.2) = 7.8.$$

Multiple choice — medium 15

34 For $f(x) = 4x^3$, find the differential dy when $x = 4$ and $dx = 0.1$.

- a. 38.4
- b. 6.4
- c. 19.2 ✓
- d. -19.2

$$dy = f'(x) dx = 12x^2 dx, \text{ so } dy = 12(4)^2(0.1) = 192(0.1) = 19.2.$$

35 For $f(x) = x^3$, find the differential dy when $x = 6$ and $dx = 0.3$.

- a. 64.8
- b. -32.4
- c. 32.4 ✓
- d. 10.8

$$dy = f'(x) dx = 3x^2 dx, \text{ so } dy = 3(6)^2(0.3) = 108(0.3) = 32.4.$$

36 For $f(x) = x^3$, find the differential dy when $x = 4$ and $dx = -0.5$.

a. -24 ✓

b. -8

c. -48

d. 24

$$dy = f'(x) dx = 3x^2 dx, \text{ so } dy = 3(4)^2(-0.5) = 48(-0.5) = -24.$$

37 For $f(x) = 2x^2$, find the differential dy when $x = 5$ and $dx = -0.1$.

a. -2 ✓

b. -1

c. 2

d. -4

$$dy = f'(x) dx = 4x^1 dx, \text{ so } dy = 4(5)^1(-0.1) = 20(-0.1) = -2.$$

38 For $f(x) = 6x^2$, find the differential dy when $x = 6$ and $dx = -0.1$.

a. -3.6

b. 7.2

c. -7.2 ✓

d. -14.4

$$dy = f'(x) dx = 12x^1 dx, \text{ so } dy = 12(6)^1(-0.1) = 72(-0.1) = -7.2.$$

39 For $f(x) = 4x^3$, find the differential dy when $x = 4$ and $dx = -0.2$.

a. -38.4 ✓

b. -12.8

c. -76.8

d. 38.4

$$dy = f'(x) dx = 12x^2 dx, \text{ so } dy = 12(4)^2(-0.2) = 192(-0.2) = -38.4.$$

40 For $f(x) = 5x^3$, find the differential dy when $x = 2$ and $dx = 0.2$.

a. 12 ✓

b. 4

c. 24

d. -12

$$dy = f'(x) dx = 15x^2 dx, \text{ so } dy = 15(2)^2(0.2) = 60(0.2) = 12.$$

41 For $f(x) = 5x^2$, find the differential dy when $x = 3$ and $dx = 0.3$.

a. 18

b. -9

c. 4.5

d. 9 ✓

$$dy = f'(x) dx = 10x dx, \text{ so } dy = 10(3)(0.3) = 30(0.3) = 9.$$

42 For $f(x) = 6x^3$, find the differential dy when $x = 6$ and $dx = -0.3$.

a. -64.8

b. 194.4

c. -194.4 ✓

d. -388.8

$$dy = f'(x) dx = 18x^2 dx, \text{ so } dy = 18(6)^2(-0.3) = 648(-0.3) = -194.4.$$

43 For $f(x) = 4x^3$, find the differential dy when $x = 2$ and $dx = 0.5$.

- a. 8
- b. -24
- c. 48
- d. 24 ✓

$$dy = f'(x) dx = 12x^2 dx, \text{ so } dy = 12(2)^2(0.5) = 48(0.5) = 24.$$

44 For $f(x) = 6x^3$, find the differential dy when $x = 2$ and $dx = -0.1$.

- a. -14.4
- b. -2.4
- c. -7.2 ✓
- d. 7.2

$$dy = f'(x) dx = 18x^2 dx, \text{ so } dy = 18(2)^2(-0.1) = 72(-0.1) = -7.2.$$

45 For $f(x) = 2x^2$, find the differential dy when $x = 4$ and $dx = 0.2$.

- a. 3.2 ✓
- b. -3.2
- c. 1.6
- d. 6.4

$$dy = f'(x) dx = 4x^1 dx, \text{ so } dy = 4(4)^1(0.2) = 16(0.2) = 3.2.$$

46 For $f(x) = 4x^3$, find the differential dy when $x = 6$ and $dx = -0.5$.

- a. 216
- b. -72
- c. -216 ✓
- d. -432

$$dy = f'(x) dx = 12x^2 dx, \text{ so } dy = 12(6)^2(-0.5) = 432(-0.5) = -216.$$

47 For $f(x) = 3x^2$, find the differential dy when $x = 1$ and $dx = 0.2$.

a. 1.2 ✓

b. 2.4

c. -1.2

d. 0.6

$$dy = f'(x) dx = 6x^1 dx, \text{ so } dy = 6(1)^1(0.2) = 6(0.2) = 1.2.$$

48 For $f(x) = 5x^3$, find the differential dy when $x = 1$ and $dx = -0.5$.

a. -2.5

b. 7.5

c. -15

d. -7.5 ✓

$$dy = f'(x) dx = 15x^2 dx, \text{ so } dy = 15(1)^2(-0.5) = 15(-0.5) = -7.5.$$

Multiple choice — hard 15

49 Evaluate $\lim_{x \rightarrow 0} \frac{e^{8x} - 1}{x}$ using L'Hôpital's Rule.

a. -8

b. 8 ✓

c. 4

d. $\frac{1}{8}$

$$\text{Form } \frac{0}{0}: \lim_{x \rightarrow 0} \frac{8e^{8x}}{1} = \frac{8e^0}{1} = \frac{8}{1} = 8.$$

50 Evaluate $\lim_{x \rightarrow 0} \frac{e^{4x} - 1}{3x}$ using L'Hôpital's Rule.

a. $\frac{4}{3}$ ✓

b. $\frac{3}{4}$

c. 1

d. $-\frac{4}{3}$

Form $\frac{0}{0}$: $\lim_{x \rightarrow 0} \frac{4e^{4x}}{3} = \frac{4e^0}{3} = \frac{4}{3} = \frac{4}{3}$.

51 Evaluate $\lim_{x \rightarrow 0} \frac{e^{3x} - 1}{2x}$ using L'Hôpital's Rule.

a. 1

b. $-\frac{3}{2}$

c. $\frac{2}{3}$

d. $\frac{3}{2}$ ✓

Form $\frac{0}{0}$: $\lim_{x \rightarrow 0} \frac{3e^{3x}}{2} = \frac{3e^0}{2} = \frac{3}{2} = \frac{3}{2}$.

52 Evaluate $\lim_{x \rightarrow 0} \frac{e^{-x} - 1}{x}$ using L'Hôpital's Rule.

a. 1

b. $-\frac{1}{2}$

c. $-\frac{3}{2}$

d. -1 ✓

Form $\frac{0}{0}$: $\lim_{x \rightarrow 0} \frac{-1e^{-x}}{1} = \frac{-1e^0}{1} = \frac{-1}{1} = -1$.

53 Evaluate $\lim_{x \rightarrow 0} \frac{e^{3x} - 1}{5x}$ using L'Hôpital's Rule.

a. $\frac{1}{2}$

b. $\frac{3}{5}$ ✓

c. $-\frac{3}{5}$

d. $\frac{5}{3}$

Form $\frac{0}{0}$: $\lim_{x \rightarrow 0} \frac{3e^{3x}}{5} = \frac{3e^0}{5} = \frac{3}{5} = \frac{3}{5}$.

54 Evaluate $\lim_{x \rightarrow \infty} \frac{5x^2 + 3x + 4}{8x^2 + 7x}$ using L'Hôpital's Rule, applied twice.

- a. $\frac{3}{4}$
- b. $\frac{5}{9}$
- c. $\frac{8}{5}$
- d. $\frac{5}{8}$ ✓

Form $\frac{\infty}{\infty}$: differentiate twice: $\lim_{x \rightarrow \infty} \frac{10}{16} = \frac{5}{8} = \frac{5}{8}$.

55 Evaluate $\lim_{x \rightarrow \infty} \frac{8x^2 - 6x - 5}{7x^2 - 2x + 8}$ using L'Hôpital's Rule, applied twice.

- a. $\frac{8}{7}$ ✓
- b. $\frac{9}{7}$
- c. $\frac{7}{8}$
- d. 1

Form $\frac{\infty}{\infty}$: differentiate twice: $\lim_{x \rightarrow \infty} \frac{16}{14} = \frac{8}{7} = \frac{8}{7}$.

56 Evaluate $\lim_{x \rightarrow \infty} \frac{2x^2 - 5}{7x^2 + x + 9}$ using L'Hôpital's Rule, applied twice.

- a. $\frac{7}{2}$
- b. $\frac{1}{4}$
- c. $\frac{3}{7}$
- d. $\frac{2}{7}$ ✓

Form $\frac{\infty}{\infty}$: differentiate twice: $\lim_{x \rightarrow \infty} \frac{4}{14} = \frac{2}{7} = \frac{2}{7}$.

57 Evaluate $\lim_{x \rightarrow \infty} \frac{2x^2 + 3x - 7}{5x^2 + 2x + 3}$ using L'Hôpital's Rule, applied twice.

- a. $\frac{3}{5}$
- b. $\frac{5}{2}$
- c. $\frac{1}{3}$
- d. $\frac{2}{5}$ ✓

Form $\frac{\infty}{\infty}$: differentiate twice: $\lim_{x \rightarrow \infty} \frac{4}{10} = \frac{2}{5} = \frac{2}{5}$.

58 Evaluate $\lim_{x \rightarrow \infty} \frac{7x^2 - 6x - 9}{3x^2 - 7x + 8}$ using L'Hôpital's Rule, applied twice.

- a. $\frac{7}{4}$
- b. $\frac{8}{3}$
- c. $\frac{7}{3}$ ✓
- d. $\frac{3}{7}$

Form $\frac{\infty}{\infty}$: differentiate twice: $\lim_{x \rightarrow \infty} \frac{14}{6} = \frac{7}{3} = \frac{7}{3}$.

59 Evaluate $\lim_{x \rightarrow 0} \frac{1 - \cos(8x)}{x^2}$ using L'Hôpital's Rule, applied twice.

- a. -32
- b. 4
- c. 16
- d. 32 ✓

Form $\frac{0}{0}$: $\lim_{x \rightarrow 0} \frac{8\sin(8x)}{2x}$, still $\frac{0}{0}$: apply again: $\lim_{x \rightarrow 0} \frac{64\cos(8x)}{2} = \frac{64}{2} = 32$.

60 Evaluate $\lim_{x \rightarrow 0} \frac{1 - \cos(2x)}{x^2}$ using L'Hôpital's Rule, applied twice.

- a. 2 ✓
- b. 1
- c. -2
- d. 3

Form $\frac{0}{0}$: $\lim_{x \rightarrow 0} \frac{2\sin(2x)}{2x}$, still $\frac{0}{0}$: apply again: $\lim_{x \rightarrow 0} \frac{4\cos(2x)}{2} = \frac{4}{2} = 2$.

61 Evaluate $\lim_{x \rightarrow 0} \frac{1 - \cos(6x)}{x^2}$ using L'Hôpital's Rule, applied twice.

a. 18 ✓

b. 3

c. 9

d. -18

Form $\frac{0}{0}$: $\lim_{x \rightarrow 0} \frac{6\sin(6x)}{2x}$, still $\frac{0}{0}$: apply again: $\lim_{x \rightarrow 0} \frac{36\cos(6x)}{2} = \frac{36}{2} = 18$.

62 Evaluate $\lim_{x \rightarrow 0} \frac{1 - \cos(x)}{x^2}$ using L'Hôpital's Rule, applied twice.

a. $\frac{1}{2}$ ✓

b. 0

c. $-\frac{1}{2}$

d. $\frac{1}{4}$

Form $\frac{0}{0}$: $\lim_{x \rightarrow 0} \frac{\sin(x)}{2x}$, still $\frac{0}{0}$: apply again: $\lim_{x \rightarrow 0} \frac{\cos(x)}{2} = \frac{1}{2} = \frac{1}{2}$.

63 Evaluate $\lim_{x \rightarrow 0} \frac{1 - \cos(5x)}{x^2}$ using L'Hôpital's Rule, applied twice.

a. $\frac{5}{2}$

b. $\frac{25}{4}$

c. $-\frac{25}{2}$

d. $\frac{25}{2}$ ✓

Form $\frac{0}{0}$: $\lim_{x \rightarrow 0} \frac{5\sin(5x)}{2x}$, still $\frac{0}{0}$: apply again: $\lim_{x \rightarrow 0} \frac{25\cos(5x)}{2} = \frac{25}{2} = \frac{25}{2}$.